DATABASE MANAGEMENT SYSTEMS LAB

Lab Report Submitted to

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for

Bachelor of Technology

in

Computer Science and Engineering

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LAB ASSIGNMENT- 2

Implementation of SQL Transactions using COMMIT, SAVEPOINT, ROLLBACK and execution of SELECT, INSERT, UPDATE, and ALTER operations.

Objective

To understand and implement SQL transactions using **COMMIT**, **SAVEPOINT**, and **ROLLBACK**, retrieve records using SELECT queries, and perform INSERT, UPDATE, and ALTER operations on tables.

Instructions

1. Open your preferred DBMS tool.
2. Create the required tables: CLASS, EMPLOYEES, and Client_master.
3. Insert suitable records into each table.
4. Execute SQL queries step by step.
5. Observe the output after each transaction operation.

TABLE STRUCTURES

CLASS Table

Attributes:

- ID (NUMBER)
- Name (VARCHAR)

Initial Records:
(1, 'Bravo')
(2, 'Alex')
(4, 'Cheng')

EMPLOYEES Table

Attributes:

- Employee_Id
- First_Name
- Last_Name
- Email
- Phone_Number
- Hire_Date

- Job_Id
- Salary
- Commission_Pct
- Manager_Id
- Department_Id

Client master Table

Attributes:

- Client No
- Name
- Address
- City
- Pincode
- State
- Bal_due

STEP 1: CREATE DATABASE

```
CREATE DATABASE DBMS_Yukti;
USE DBMS_Yukti;
```

STEP 2: CREATE TABLES

CLASS TABLE

```
CREATE TABLE CLASS (
  ID INT PRIMARY KEY,
  Name VARCHAR(50)
);
```

Insert initial records:

```
INSERT INTO CLASS VALUES (1,'Bravo');
INSERT INTO CLASS VALUES (2,'Alex');
INSERT INTO CLASS VALUES (4,'Cheng');
COMMIT;
```

SELECT * FROM CLASS;

	ID	Name
▶	1	Bravo
	2	Alex
	4	Cheng
*	NULL	NULL

EMPLOYEES TABLE

```
CREATE TABLE EMPLOYEES (
  Employee_Id INT PRIMARY KEY,
  First_Name VARCHAR(50),
  Last_Name VARCHAR(50),
  Email VARCHAR(50),
  Phone_Number VARCHAR(20),
  Hire_Date DATE,
  Job_Id VARCHAR(20),
  Salary DECIMAL(8,2),
  Commission_Pct DECIMAL(4,2),
  Manager_Id INT,
  Department_Id INT
);
```

Insert records:

```
INSERT INTO EMPLOYEES VALUES
(101,'John','Austin','john@gmail.com','9876543210',CURDATE(),'IT_PROG',5000,NULL,100,60),
(102,'Neha','Sharma','neha@gmail.com','9876543211',CURDATE(),'HR_REP',4800,NULL,100,70)

,
(103,'Raj','Kumar','raj@gmail.com','9876543212',CURDATE(),'IT_PROG',6000,NULL,101,80),
(104,'Sara','AUSTIN','sara@gmail.com','9876543213',CURDATE(),'SA_REP',4500,NULL,102,60),
(105,'Aman','Verma','aman@gmail.com','9876543214',CURDATE(),'SA_REP',5200,NULL,100,70
);

COMMIT;

SELECT * FROM EMPLOYEES;
```

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
101	John	Austin	john@gmail.com	9876543210	2026-03-02	IT_PROG	5000.00	NULL	100	60
102	Neha	Sharma	neha@gmail.com	9876543211	2026-03-02	HR_REP	4800.00	NULL	100	70
103	Raj	Kumar	raj@gmail.com	9876543212	2026-03-02	IT_PROG	6000.00	NULL	101	80
104	Sara	AUSTIN	sara@gmail.com	9876543213	2026-03-02	SA_REP	4500.00	NULL	102	60
105	Yukti	Vadehra	yukti@gmail.com	9876543214	2026-03-02	SA_REP	5200.00	NULL	100	70
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Client_master TABLE

```
CREATE TABLE Client_master (
  ClientNo VARCHAR(6) PRIMARY KEY,
  Name VARCHAR(50),
  Address VARCHAR(100),
  City VARCHAR(50),
  Pincode INT,
  State VARCHAR(50),
  Bal_due DECIMAL(10,2)
);
```

Insert records:

```
INSERT INTO Client_master VALUES
('C123','Rahul','MG Road','Mumbai',400001,'Maharashtra',6000),
('C124','Neha','Brigade Road','Bangalore',560001,'Karnataka',3000),
('C125','Arjun','Park Street','Kolkata',700016,'West Bengal',7000),
('C126','Meena','Anna Salai','Chennai',600002,'Tamil Nadu',2000),
('C127','Ravi','Banjara Hills','Hyderabad',500034,'Telangana',8000);

COMMIT;

SELECT * FROM Client_master;
```

ClientNo	Name	Address	City	Pincode	State	Bal_due
C123	Rahul	MG Road	Mumbai	400001	Maharashtra	6000.00
C124	Neha	Brigade Road	Bangalore	560001	Karnataka	3000.00
C125	Arjun	Park Street	Kolkata	700016	West Bengal	7000.00
C126	Meena	Anna Salai	Chennai	600002	Tamil Nadu	2000.00
C127	Ravi	Banjara Hills	Hyderabad	500034	Telangana	8000.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL

DDL OPERATIONS

Rename Client_master

```
RENAME TABLE Client_master TO Client12;
```

Display bal_due as BALANCE

```
SELECT Bal_due AS BALANCE
FROM Client12;
```

BALANCE
6000.00
3000.00
7000.00
2000.00
8000.00

PART A: TRANSACTIONS (CLASS TABLE)

```
INSERT INTO CLASS VALUES (5,'Rahul');
COMMIT;

UPDATE CLASS SET Name='Abhijeet' WHERE ID=5;
SAVEPOINT A;

INSERT INTO CLASS VALUES (6,'Chris');
SAVEPOINT B;
```

INSERT INTO CLASS VALUES (7,'Bravo');
SAVEPOINT C;

SELECT * FROM CLASS;

	ID	Name
▶	1	Bravo
	2	Alex
	4	Cheng
	5	Abhijeet
	6	Chris
	7	Bravo
•	NULL	NULL

ROLLBACK TO B

ROLLBACK TO B;
SELECT * FROM CLASS;

Output:

1	Bravo
2	Alex
4	Cheng
5	Abhijeet
6	Chris

(ID 7 removed)

ROLLBACK TO A

ROLLBACK TO A;
SELECT * FROM CLASS;

Output:

1	Bravo
2	Alex
4	Cheng
5	Abhijeet

(ID 6 also removed)

PART B: SELECT QUERIES (EMPLOYEES)

1 Employee ID, Names and Salaries

```
SELECT Employee_Id, First_Name, Last_Name, Salary
FROM EMPLOYEES;
```

Employee_Id	First_Name	Last_Name	Salary
101	John	Austin	5000.00
102	Neha	Sharma	4800.00
103	Raj	Kumar	6000.00
104	Sara	AUSTIN	4500.00
105	Yukti	Vadehra	5200.00
NULL	NULL	NULL	NULL

---

2 Employees under Manager 100

```
SELECT * FROM EMPLOYEES
WHERE Manager_Id = 100;
```

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
101	John	Austin	john@gmail.com	9876543210	2026-03-02	IT_PROG	5000.00	NULL	100	60
102	Neha	Sharma	neha@gmail.com	9876543211	2026-03-02	HR_REP	4800.00	NULL	100	70
105	Yukti	Vadehra	yukti@gmail.com	9876543214	2026-03-02	SA_REP	5200.00	NULL	100	70
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

---

3 Salary ≥ 4800

```
SELECT First_Name, Salary
FROM EMPLOYEES
WHERE Salary >= 4800;
```

First_Name	Salary
John	5000.00
Neha	4800.00
Raj	6000.00
Yukti	5200.00

---

4 Last Name = 'AUSTIN'

```
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Last_Name = 'AUSTIN';
```

	First_Name	Last_Name
▶	John	Austin
	Sara	AUSTIN

---

5 Departments 60,70,80

SELECT First_Name, Department_Id
FROM EMPLOYEES
WHERE Department_Id IN (60,70,80);

	First_Name	Department_Id
▶	John	60
	Neha	70
	Raj	80
	Sara	60
	Yukti	70

---

6 Unique Manager_Id

SELECT DISTINCT Manager_Id
FROM EMPLOYEES;

	Manager_Id
▶	100
	101
	102

PART C: UPDATE OPERATIONS (Client12)

Clients with bal_due > 5000

SELECT Name
FROM Client12
WHERE Bal_due > 5000;

	Name
▶	Rahul
	Arjun
	Ravi

Update bal_due of C123

```
UPDATE Client12
SET Bal_due = 5100
WHERE ClientNo = 'C123';

COMMIT;
SELECT * FROM Client12 WHERE ClientNo='C123';
```

	ClientNo	Name	Address	City	Pincode	State	Bal_due
▶	C123	Rahul	MG Road	Mumbai	400001	Maharashtra	5100.00
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL

RESULT

Thus, SQL transactions using COMMIT, SAVEPOINT, and ROLLBACK were successfully executed. Records were inserted, retrieved, and updated successfully.